A Comparative Study of 3D Input Architectures for Neural Network Predictions of MeV Proton Scattering

Background: Accurately modeling the scattering of MeV protons through complex geometries is necessary for several fields. For nuclear applications, data is used to predict radiation shielding needed in nuclear reactors or high-energy environments like space-based hardware that experiences high background radiation. Similarly, in the medical sector, the data is needed in proton radiography for cancer patients.

Proposed Research: I'm proposing a project that inputs 3D geometries and structural data (e.g. density) to predict the resulting proton angular distribution and energy spectrum of the structure after being hit by 100 MeV protons. I will compare 3 input methods for the geometry: 3D voxel grids (3D-CNN), point clouds (PointNet), and GNN's (point-GNN/point-voxel CNN). I will train on data using Geant4 (particle physics simulations) data where I input simple geometries like an aluminum slab. My loss function will be a linear combination of the angular and energy loss. After determining the best model for this problem, I'll find an improvement such as through a more involved loss term or a model tweak.

Sources:

https://geant4.web.cern.ch/

https://www.thinkautonomous.ai/blog/voxel-vs-points/#point-based-approaches-from-pointnet-to-point-transformers

3D-CNN - https://www.sciencedirect.com/science/article/pii/S0927025620303414

PointNet - https://arxiv.org/abs/1612.00593

Point-Voxel - https://arxiv.org/abs/1907.03739